

P2 - CSG - Sheet and Print

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Revision: 2026-09-23. Instructor review draft. Grading allocation approved; dates remain proposed for review before publication.

2.1 - CSG process and instructions

Introduced: Week 2 - Sep 1 / **Due:** Week 6 - Sep 29, 11:59 PM Eastern Time / **Weight:** 10%

Description

Build an editable CSG object and explain its construction as a sequence another person can follow. Use a BIG-style massing diagram: one operation per frame, a consistent viewpoint and a visible change between steps. Present one diagram sequence and one matching set of instructions (pseudocode) on the same 17 x 11 inch sheet, then make one 3D printed object in P2b.

Rules & Constraints

- One 17 x 11 inch sheet laid out as a BIG-style process diagram (in the manner of Bjarke Ingels Group's step-by-step massing diagrams): six to nine numbered frames, according to the actual operation sequence, with one clear operation or operation type per frame.
- Use white building masses, color to identify operations or changes, clear outlines and a simple line-weight hierarchy. Use lighter or dashed lines for hidden edges where needed. Keep the viewpoint consistent; introduce context and vegetation in the final frame. Follow your Graphic Standards Manual.
- Use at least three primitives; include translation, rotation and scale, plus union and difference. Intersection is demonstrated in class and optional in the assignment. Label the kept solid and cutter in each subtraction.
- Draw the final solid at building scale with a 1.75 m figure, vegetation, ground and shadow.
- Add clear instructions and matching pseudocode beside the diagrams, one step per operation, with the exposed parameters.
- Refer to Di Mari and Yoo's Operative Design and Di Mari's Conditional Design; name the spatial operation your sequence explores.
- Submit an editable graph with exposed inputs and all source operations intact.

Parameters

- Expose primitive dimensions, translation values, rotation angle/axis/center and scale factor. Name inputs by their geometric role and state units.
- Define the origin and axes, where each primitive starts, and the order of Boolean operands. In a difference, identify the solid kept and the cutter removed.
- Keep the source operations editable. Preserve native node names and use Frames/groups to explain the sequence. Use Blender Geometry Nodes or an already approved Rhino/Grasshopper route.

Required Studies

- **Six to nine numbered frames.** Begin with at least three primitive solids. Include translation, rotation, scale, union and difference; intersection is optional. Mark incoming geometry and removed material so the change can be read without guessing. Each frame records one operation or operation type, with its relevant values.
- **Matching instructions (pseudocode).** Place one instruction beside each frame. Identify objects by primitive type, position or step number; add consistent names when the sequence becomes complex enough to need them. State units and coordinate conventions once, then specify enough dimensions, translations and other transforms to reconstruct the result. Naming every intermediate solid is optional.

- **BIG-style graphic language.** Keep building masses white and use your palette to highlight added, removed or transformed geometry. Use strong outer outlines, lighter secondary edges and dashed hidden edges where useful. Study the [diagram tutorial](#), also listed in Resources. Reserve vegetation and other context for the final frame.
- **Final solid at building scale.** Show the final geometry with a 1.75 m figure, vegetation, ground and shadow. Separate presentation context from the CSG object that will be fabricated. State the spatial operation explored, with a reference to Operative Design or Conditional Design.

Workflow

1. Choose a spatial operation from the readings, sketch a sequence and build it from primitives using an editable graph.
2. Check the Boolean order and change an exposed input. Verify that the downstream result recomputes without rebuilding the graph.
3. Write the procedure from the working model. Ask a partner to identify any undefined size, position or operand; revise those instructions.
4. Export the operation views from one camera and compose the single sheet using the Graphic Standards Manual. Match every frame to the current graph.

Deliverables

- **One 17 x 11 inch landscape PDF sheet:** Lastname_Firstname_P2a.pdf. Include all numbered frames, adjacent instructions/pseudocode, exposed values, the final architectural view and source credits.
- **Editable graph:** .blend, or .gh plus .3dm where used. Internalize Grasshopper reference inputs; package any other necessary dependencies. A baked mesh or screenshots alone cannot demonstrate the operation history.
- Use Lastname_Firstname_P2a/ for the submission. Keep the object available for the separate P2b fabrication milestone.

Evaluation

- **Computational understanding - 30%:** explain the transforms, Boolean operands and operation order; demonstrate how an exposed input changes the result.
- **Visual communication - 30%:** diagram and pseudocode agree; consistent views explain each change and the final spatial result using the Graphic Standards Manual.
- **Technical execution - 25%:** required operations remain editable; dimensions, axes, operands and instructions allow reconstruction; the submission files are complete.
- **Exploration - 15%:** purposeful parameter tests and the chosen operation sequence support the intended spatial effect.

Before You Submit

- [] Six to nine frames; at least three primitives; translation, rotation, scale, union and difference. Intersection is optional.
- [] Difference frames label the kept solid and cutter.
- [] Instructions and pseudocode match the graph and include parameter values.
- [] Final view includes a 1.75 m figure, vegetation, ground and shadow.
- [] One sheet and an editable source that opens with all dependencies.

2.2 - CSG 3D printed object

Introduced: Week 3 - Sep 8 / **Due:** Week 8 - Oct 13, 11:59 PM Eastern Time / **Weight:** 5%

Preparation checks: Week 6 - Sep 29: Check geometry, units, orientation and the sliced preview before queueing.

Description

Fabricate a variant of your P2a CSG object and explain the decisions that make it printable. Compare the final physical object with the digital geometry. Record any revision for closure, feature thickness or orientation so the fabrication copy remains traceable to the original construction.

Rules & Constraints

- Print a variant of your CSG object from P2a, within 60 mm on each axis and the 200 g allocation, without supports.
- Hand in the physical print during class on the listed due date; submit digital files and brief fabrication notes by 11:59 PM Eastern Time that night. No additional presentation sheet is required.
- Plan how to avoid support material through orientation, separating the object into parts for assembly, or adjusting overhangs. Bridging is allowed when it prints successfully on the selected printer.
- Submit the editable source, exported mesh and slicer project. Explain any geometry changes made for printability.

Parameters

- Uniform print scale and final X/Y/Z dimensions in millimeters. Each axis must be no greater than 60 mm for the final assembled object; confirm each part also fits within 60 mm on each axis for printing.
- Printer/material profile, draft layer height, wall count and infill. Supports must be off. Record the chosen values rather than assuming a universal printer profile.
- Orientation, bed-contact face, estimated time and filament use. The total allocation is 200 g including failed attempts.

Required Studies

- **Geometry check.** Show the final fabrication mesh as a closed solid with consistent outward normals. Resolve open edges, internal faces and self-intersections before slicing.
- **Orientation comparison.** Compare two positions on the bed. Label contact areas, overhangs, bridges and unsupported islands; explain the chosen orientation. Separate the object into parts and assemble after printing, or adjust overhangs where needed. Bridging is acceptable if it works with the chosen printer and material; check the sliced preview and test uncertain spans.
- **Sliced and physical result.** Inspect the layer preview for missing thin features and floating starts. Record dimensions, settings, predicted time/material and a photograph of the completed object. Explain one difference between the prediction and the print.

Workflow

1. Duplicate/export the P2a result for fabrication while keeping its source graph intact. Record changes made to the geometry.
2. Export the mesh in millimeters and confirm all three dimensions in the slicer; STL itself does not store units. Scale uniformly to preserve proportions.
3. Complete the geometry/orientation/slicing check before entering the queue. Follow current FabLab orientation and machine procedures.
4. Print with supports off, keep a record of material used including failed attempts, and photograph the final object, assembling printed parts if needed. Include brief notes explaining your support-free strategy.

Deliverables

- **One physical CSG object**, assembled if printed in parts, within 60 x 60 x 60 mm, produced without supports and within the 200 g allocation. Bring it to class on the listed due date with your name on a separate label.
- **Brief fabrication notes and a photograph** in the submission folder. Record your support-free strategy, orientation choice, any part separation/assembly or geometry changes, dimensions and slicer settings/time/material. The 2.1 diagram sheet remains the single CSG presentation sheet; no second sheet is required.
- **Editable source, exported mesh and slicer project** in Lastname_Firstname_P2b/. Include a screenshot showing dimensions and supports off. A preview or render does not replace the physical object.

Evaluation

- **Preparation and fabrication strategy - 30%:** mesh, orientation, part division and sliced layers support a viable print within the size, material and support constraints.
- **Fabrication quality - 30%:** the completed object retains the intended geometry, with sound features, controlled finish and accurate assembly where printed in parts.
- **Visual documentation - 25%:** clear photographs and notes follow the Graphic Standards Manual; the source, mesh, slicer project and physical result correspond.
- **Testing and revision - 15%:** orientation comparisons, layer checks and any print tests inform decisions; explain what worked and any changes made.

Before You Submit

- [] Each dimension is at most 60 mm; supports are off; cumulative filament use is at most 200 g.
- [] Mesh closure, normals, thin features and layer preview have been checked.
- [] Brief fabrication notes include the support-free strategy, orientation reasoning, settings and a photograph.
- [] Physical object, source, mesh and slicer project are ready for submission.

Shared standards: Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages follow the instructor review draft. This handout uses US Letter; student sheets are 17 x 11 inches.

Midterm grade: The midterm grade accumulates the Graphic Standards Manual, CSG sheet and print, module (3.1) and arrays (3.2). These assignments account for 40% of the final course grade. Report the accumulated result as a percentage of those 40 possible course points. The midterm adds no separate assessment weight.